

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F\,D$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$-\overline{N_{j_1,r_1}}\,N_{j_1,r_1}\,M_F$	$-N_0\,\text{wfr}\,\text{color1}_{1q}\,\text{NewStr}\,\text{color1}_{1q}^\dagger\,H_U^0$	$-M_U^0\,N_0\,\text{wfr}\,\text{color1}_{1q}\,N_0\,\text{wfr}\,\text{color1}_{1q}^\dagger$	$+i\overline{N_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\,(N_{j_1,r_1})$
									$+\frac{1}{2}\text{amb}\,\text{ds1}\,\text{Inv1}\,N_0\,\text{wfr}\,\text{color1}_{1q}\,\text{NewStr}\,\text{color1}_{1q}\,\text{T1}\,\text{Str}\,\text{color1}_{1q}\,\text{HC}\,\text{GInv1}_{1q,1q,1q,1q}\,\text{NewStr}\,\text{color1}_{1q}^\dagger\,\text{NewStr}\,\text{color1}_{1q}$	$+\frac{1}{2}\text{amb}\,\text{ds1}\,\text{Inv1}\,N_0\,\text{wfr}\,\text{color1}_{1q}\,\text{NewStr}\,\text{color1}_{1q}\,\text{T1}\,\text{Str}\,\text{color1}_{1q}\,\text{HC}\,\text{GInv1}_{1q,1q,1q,1q}\,\text{NewStr}\,\text{color1}_{1q}^\dagger\,\text{NewStr}\,\text{color1}_{1q}$	
									$+\frac{1}{2}\text{amb}\,\text{ds1}\,\text{Inv1}\,N_0\,\text{wfr}\,\text{color1}_{1q}\,\text{NewStr}\,\text{color1}_{1q}\,\text{T1}\,\text{Str}\,\text{color1}_{1q}\,\text{HC}\,\text{GInv1}_{1q,1q,1q,1q}\,\text{NewStr}\,\text{color1}_{1q}^\dagger\,\text{NewStr}\,\text{color1}_{1q}$	$+\frac{1}{2}\text{amb}\,\text{ds1}\,\text{Inv1}\,N_0\,\text{wfr}\,\text{color1}_{1q}\,\text{NewStr}\,\text{color1}_{1q}\,\text{T1}\,\text{Str}\,\text{color1}_{1q}\,\text{HC}\,\text{GInv1}_{1q,1q,1q,1q}\,\text{NewStr}\,\text{color1}_{1q}^\dagger\,\text{NewStr}\,\text{color1}_{1q}$	

Field combinations 2 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_1	H	S_1	S_2	S_2	D
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger$	D_{μ_1}	$(\text{NewScalar1}_{j_1})^\dagger$	$+D_{\mu_1}$	$(\text{NewScalar2}_{j_1})^\dagger$	D_{μ_1}	$(\text{NewScalar2}_{j_1})^\dagger$

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	+iam bdaH2Inv1 H_{a_2}NewScalar2$_{j_2}$T3Hgg1Partial2CGIw1$_{a_1,a_2,j_1,j_2}$$H_{a_1}$NewScalar2$_{j_1}$[†] +iam bdaH2Inv2 H_{a_2}NewScalar2$_{j_2}$T3Hgg1Partial2CGIw2$_{a_1,a_2,j_1,j_2}$$H_{a_1}$NewScalar2$_{j_1}$[†]	$+ \overline{\ell_{a_1,i_1}} C P_R \overline{N_{j_1,r_1}}^T$NewScalar1$_{j_1}$T3Y1CG$_{a_1,j_1,j_1} y^{1,i_1,r_1}$ $+ \overline{y^{1,i_1,r_1}} \ell_{a_1,i_1}^T C P_L N_{j_1,r_1}$T3Y1CG$_{a_1,j_1,j_1}$NewScalar1$_{j_1}$[†]	$+ \overline{\lambda_{\tau 2}}$NewScalar2$_{j_1}$T3MixCG$_{a_1,a_2,j_1,j_1} H_{a_1}^\dagger H_{a_2}^\dagger$NewScalar1$_{j_1}$[†] $+ H_{a_1} H_{a_2}$NewScalar1$_{j_1}$T3MixCG$_{a_1,a_2,j_1,j_1} \lambda_{\tau 2}$NewScalar2$_{j_1}$[†]	$+ \overline{y^{2,i_1,r_1}} \overline{N_{j_1,r_1}} P_L \ell_{a_1,i_1}$NewScalar2$_{j_1}$T3Y2CG$_{a_1,j_1,j_1}$ $+ \overline{\ell_{a_1,i_1}} P_R N_{j_1,r_1}$T3Y2CG$_{a_1,j_1,j_1} y^{2,i_1,r_1}$NewScalar2$_{j_1}$[†]